Commutative Algebra Problem Set #6

Dmytro Zakharov

Problem (Exercise 1)**.** *Let $Z = Spec(A) \setminus \{\eta\}$ where η is the generic point. Prove that there does not exist a finitely generated $\mathbb{Z}$-algebra A such that the image of $Spec(A) \to Spec(\mathbb{Z})$ is equal to Z.*

Solution. Suppose otherwise that there exists some finitely generated $\mathbb{Z}$-algebra $A \cong \mathbb{Z}[T_1, \dots, T_n]/I$ such that $f : \mathrm{Spec}(A) \to \mathrm{Spec}(\mathbb{Z})$ has the image of precisely Z. Note that by the problem statement, $f^{-1}(\eta) = \emptyset = \mathrm{Spec}(A \otimes_{\mathbb{Z}} \kappa(0))$. Note:

$$\begin{aligned}\mathrm{Spec}(A \otimes_{\mathbb{Z}} \kappa(0)) &= \mathrm{Spec}(\mathbb{Z}[T_1, \dots, T_n]/I \otimes_{\mathbb{Z}} \mathbb{Q}) \\ &= \mathrm{Spec}(\mathbb{Q}[T_1, \dots, T_n]/(I\mathbb{Q}[T_1, \dots, T_n])) = \emptyset\end{aligned}$$

Thus, we obtain $\mathbb{Q}[T_1, \dots, T_n]/(I\mathbb{Q}[T_1, \dots, T_n]) \cong 0$. This can only happen in the following scenario:

$$\mathbb{Q}[T_1, \dots, T_n] = I\mathbb{Q}[T_1, \dots, T_n], \quad I \subseteq \mathbb{Z}[T_1, \dots, T_n]$$

We claim that this implies that there exists a non-zero integer $N \in I$. Indeed, since $1 \in I\mathbb{Q}[T_1, \dots, T_n]$, there exist $a_1/b_1, \dots, a_r/b_r \in \mathbb{Q}$ and $f_1, \dots, f_r \in I$ such that $\sum_{i=1}^{r} (a_i/b_i) f_i = 1$. Multiply both sides by $N := \prod_{i=1}^{r} b_i$ to get $\mathbb{Z}$-linear combination of $f_1, \dots, f_r \in I$ that equals $N \neq 0$. Now, note:

$$f(\mathfrak{p}) = \mathbb{Z} \cap \mathfrak{p} \subseteq \{p\mathbb{Z} : p \mid N\}.$$

Finally, $f^{-1}(q\mathbb{Z}) = \emptyset$ for any prime $q > N$ — contradiction. □

Problem (Exercise 2)**.** *Let $k = \mathbb{F}_q$ be the field with q elements and let*

$$A = k[X, Y]/(X^q Y - X Y^q)$$

Let $a, b \in k$ not both zero, consider the k-algebra map

$$\phi_{a,b} : k[T] \to A, \; T \mapsto aX + bY.$$

Prove that the map $\phi_{a,b}$ is not finite.

Solution. Suppose otherwise that the map $\phi_{a,b}$ is finite. That said, for fixed $e_1, \dots, e_n \in A$, any element $f(X, Y) \in A$ can be written as:

$$f(X, Y) = \sum_{i=1}^{n} c_i(aX + bY) e_i(X, Y), \quad \text{for some } c_1, \dots, c_n \in k[T]$$

Suppose $d := \max_{i\in[n]}\{\deg e_i\}$, then set $f(X,Y) := Y^{d+1}$. Thus we have:

$$Y^{d+1} = \sum_{i=1}^{n} c_i(aX+bY)e_i(X,Y)$$

If this equality holds, then it should hold for any substitution. Thus, consider $X := -(b/a)Y$. Then, we have:

$$Y^{d+1} = \sum_{i=1}^{n} c_i(0)e_i(-bY/a,Y) = \sum_{i=1}^{n} \gamma_i \widetilde{e}_i(Y).$$

Degree of the left-hand side is $d+1$, while the degree of right-hand side is bounded above by d. Contradiction. □

Problem (Exercise 3)**.** *Let k be a field and A a finitely generated domain over k. Let L be the integral closure of k in A. Show that L is finite over k.*

Solution. As noted in the hint, we show that L is a field. Suppose $\ell \in L$, thus ℓ is the root of some monic polynomial

$$\ell^n + c_1\ell^{n-1} + \cdots + c_n = 0, \quad c_j \in k.$$

Thus, we have:

$$\ell(\ell^{n-1} + \cdots + c_{n-1}) = -c_n \Rightarrow \ell^{-1} = -\frac{1}{c_n}\ell^{n-1} - \frac{c_1}{c_n}\ell^{n-2} - \cdots - \frac{c_{n-1}}{c_n}.$$

Note that we can safely assume $c_n \neq 0$ (or otherwise $\ell = 0$ which surely does not have an inverse ℓ^{-1}). While we have successfully found $\ell^{-1} \in k$ explicitly, why is it in L? First note that $\ell^{-1} \in A$ because each $\ell^j \in A$ and on the right-hand side we have a k-linear combination of elements of A. Second, ℓ^{-1} is clearly the root of polynomial

$$c_n(\ell^{-1})^n + c_{n-1}(\ell^{-1})^{n-1} + \cdots + c_1\ell^{-1} + 1 = 0$$

Thus $\ell^{-1} \in L$. Therefore, L is a field.

Now, take $\mathfrak{m} \subseteq A$ some maximal ideal. From Nullstellensatz, $A/\mathfrak{m}$ is a finite field extension of k. Define $\pi : L \hookrightarrow A/\mathfrak{m}$ — a natural restriction of $A \to A/\mathfrak{m}$ over L. Note that this is a valid embedding: it is clearly a homomorphism, and moreover since L is a field, $\ker(\pi)$ is either (0) or (1), the latter being clearly impossible since $\pi(1) = 1$. Thus:

$$L \cong \pi(L) \subseteq A/\mathfrak{m}.$$

Since $[A/\mathfrak{m} : k] < \infty$, $[L : k] < \infty$, and so L is finite over k. □

Problem (Exercise 4)**.** *Let X be a spectral space and let $W \subseteq X$ be a constructible subset. Prove that W is quasi-compact.*

Solution. Since W is constructible, $W = \bigcup_{\alpha \in I} U_\alpha \cap V_\alpha$ for I — finite, and U_α are open while V_α are closed.

Now suppose $W = \bigcup_{\gamma \in J} W_\gamma$ for open W_γ and potentially infinite J. Therefore, $\bigcup_{\gamma \in J} W_\gamma = \bigcup_{\alpha \in I} U_\alpha \cap V_\alpha$. Fix some α. V_α is quasi-compact since it is closed. Then, $\{V_\alpha \cap W_\gamma\}_{\gamma \in J}$ covers $V_\alpha \cap U_\alpha$. Since V_α is quasi-compact, the open cover $\{V_\alpha \cap W_\gamma\}_{\gamma \in J}$ has a finite subcover (say, indexed by finite J_α). Thus, we can construct the finite subcover as:

$$W = \bigcup_{\alpha \in I} \bigcup_{\gamma \in J_\alpha} W_\gamma$$

with $\sum_{\alpha \in I} |J_\alpha| < \infty$. □

Problem (Exercise 5)**.** *Let A be a ring and let $Z \subseteq \mathrm{Spec}(A)$ be a closed subset. Prove that Z is constructible if and only if there exists a finitely generated ideal $I \subseteq A$ such that $Z = V(I)$.*

Solution. Further denote $X := \mathrm{Spec}(A)$.

($\Rightarrow$) Assume Z is constructible, thus quasi-compact by *Exercise 4*. Then, $X \setminus Z = \bigcup_{i=1}^{n} D(f_i)$. Take complement to get $Z = \bigcap_{i \in I} V(f_i) = V(f_1, \dots, f_n)$. Thus $I := (f_1, \dots, f_n)$ is the needed finitely generated ideal.

($\Leftarrow$) Let $I = (f_1, \dots, f_n)$ for some $f_i \in A$. Then, $Z = V(I) \cap D(1)$. □

Problem (Exercise 6*)**.** *Let $Z = \mathrm{Spec}(A) \setminus \{\eta\}$ where η is the generic point. Prove that there does not exist a $\mathbb{Z}$-algebra A such that the image of $\mathrm{Spec}(A) \to \mathrm{Spec}(\mathbb{Z})$ is equal to Z.*

Proof (I guess?) Similarly to *Exercise 1*, assuming that such $\mathbb{Z}$-algebra exists, $A \otimes_{\mathbb{Z}} \mathbb{Q} = 0$. This more or less evidently implies that A is a torsion $\mathbb{Z}$-module, so in particular $n\mathbf{1}_A = 0$ and thus moreover $\mathrm{char}(A) = n$. This similarly requires image to be the subset of $\{p\mathbb{Z} : p \mid n\}$, which is finite, thus the contradiction. □